

JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

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CLASS

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BIOLOGY

9744/04

Paper 4 Practical

15 August 2017

2 hours 30 minutes

Candidates answer on the Question Paper.

Additional Materials: As listed in the Confidential Instructions.

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Give details of the practical shift and laboratory, where appropriate, in the boxes provided.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	
2	
3	
Total	

This document consists of **17** printed pages.

[Turn over

Answer **all** questions.

- 1 During the light dependent stage of photosynthesis, hydrogen ions and electrons are transferred to hydrogen acceptor molecules, including NADP.

DCPIP (2,6-dichlorophenolindophenol) is a blue dye, which acts as a hydrogen ion and electron acceptor. As DCPIP accepts hydrogen ions or electrons it is reduced and becomes colourless.

You are required to investigate the effect of different wavelengths of light on the rate of the light dependent stage of photosynthesis in a leaf extract containing chloroplasts.

You are provided with:

- a leaf extract in buffered solution, labelled **L**, in a beaker with ice,
- DCPIP solution, labelled **D**,
- filters that allow light of specific wavelengths to pass through, as shown in Table 1.1.

Table 1.1

colour	label	wavelength / nm
purple	P	425
blue	B	450
green	G	525
orange	O	625
red	R	675

The leaf extract is an irritant. It is recommended that you wear safety goggles/glasses and gloves.

Proceed as follows.

1. Stir the leaf extract, **L**, using the glass rod.
2. Use a syringe to draw up 0.5 cm³ of **L**.
3. Wipe the outside of the syringe to remove any liquid.
4. Put the syringe at the centre of a white tile. This will be used as a colour standard.
5. You are required to add enough DCPIP solution, **D**, to change the colour of the remaining leaf extract, **L**. The change in colour must be sufficient to be observable in the 0.5 cm³ sample transferred to a syringe in step 8.
 - Using a Pasteur pipette, put about 0.5 cm³ of DCPIP solution, **D**, into the remaining leaf extract, **L**, in the specimen tube.
 - Shake the specimen tube gently so that the colour spreads evenly.
 - Tilt the specimen tube and view the colour against a white background.
 - If there is no noticeable colour change, add DCPIP solution, **D**, drop by drop until a noticeable colour change is achieved.

6. Immediately wrap the specimen tube containing the mixture of **L** and **D** in foil. Cover the specimen tube with a foil lid, as shown in Fig. 1.1. This should be easy to remove to obtain the mixture of **L** and **D**. Put back the covered specimen tube containing the mixture of **L** and **D** in the beaker with ice.

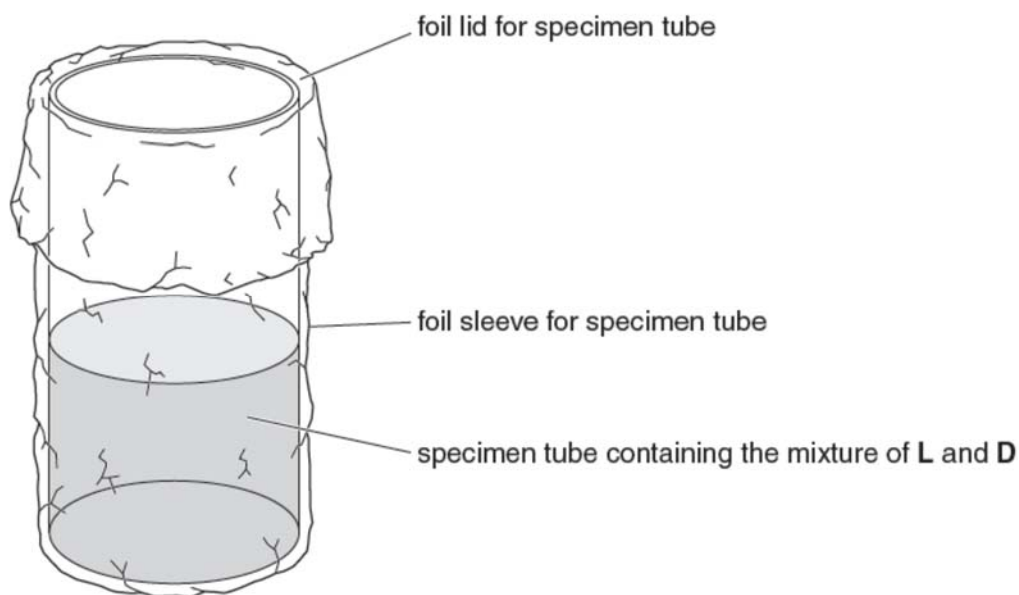


Fig. 1.1

7. Place the bench lamp 10 cm from the syringe on the white tile. Do **not** switch the lamp on.

The next steps have to be carried out very quickly one after another, so read steps 8–15 and refer to Fig. 1.2 before proceeding.

8. Remove the foil lid and use a clean syringe to draw up 0.5 cm³ of the mixture of **L** and **D** in the specimen tube. Replace the foil lid immediately.
9. Wipe the outside of the syringe and place it next to the colour standard on the white tile. This is the test syringe.
10. Immediately cover both syringes with the purple filter, **P**, as shown in Fig. 1.2 on page 4.

11. Switch on the bench lamp and immediately start a stopwatch or stop clock.

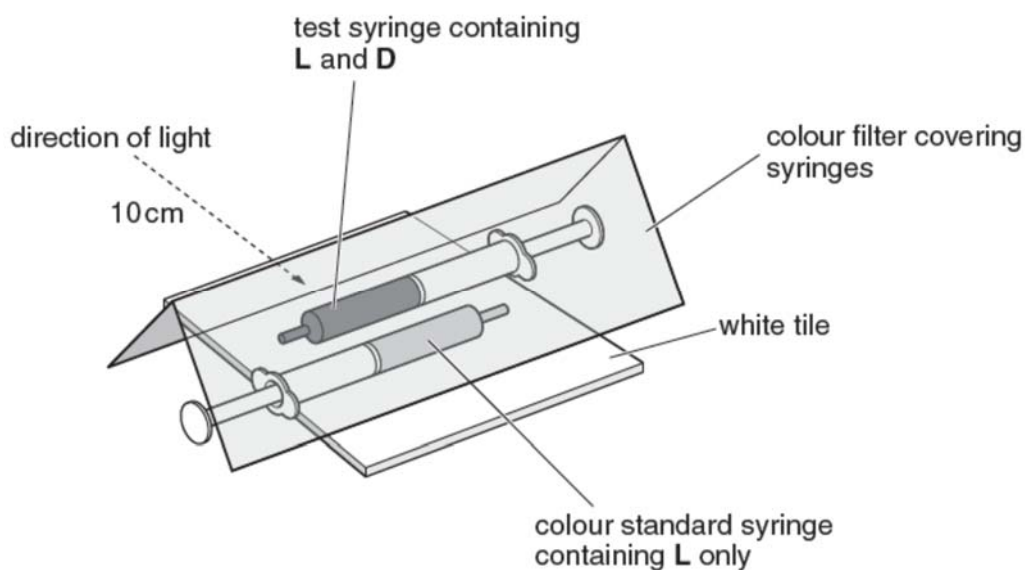


Fig. 1.2

12. In the space provided in **(a)**, record the time taken for the colour in the test syringe to match that of the colour standard. If the colour does not match after 300 seconds then record 'more than 300'.

13. Switch off the bench lamp.

14. Expel the contents of the test syringe into the beaker labelled **waste**. Rinse the syringe.

15. Repeat steps **8–14** using each of the four remaining coloured filters in turn (blue, green, orange and red).

(a) Record these results in a suitable table in the space provided to show the effect of wavelength on the time to decolourise DCPIP. [3]

(b) (i) Give one reason to explain why the leaf extract was kept on ice. [1]

(ii) State why the leaf extract containing DCPIP was kept covered by foil. [1]

(iii) Describe a suitable control that could have been set up for this investigation. [1]

(c) Suggest why the rate of photosynthesis is different at different wavelengths. [2]

(d) Suggest two significant sources of error in this experiment and describe two corresponding improvements that could be made to reduce the effects of these errors. [4]

In another similar investigation, a student collected leaves from two varieties of the same species of a garden plant that has different coloured leaves.

Variety A dark red leaves

Variety B green and white striped leaves

The student made a chloroplast extract from the leaves of each variety and measured the rate of photosynthesis for each extract in different wavelengths of light.

Table 1.2 shows the rates of photosynthesis calculated by the student from her results.

Table 1.2

wavelength of light / nm	rate of photosynthesis / s ⁻¹	
	source of chloroplasts	
	dark red leaf	green and white striped leaf
425	0.097	0.081
450	0.071	0.063
525	0.023	0.023
625	0.030	0.039
675	0.057	0.058

(e) Use the grid provided to plot line graphs showing the effect of wavelength on the rate of photosynthesis. [4]



(f) Use your graph to estimate the rate of photosynthesis at a wavelength of 430 nm in plants with dark red leaves. [1]

(g) Explain how light of wavelength 430 nm leads to the decolourisation of DCPIP. [1]

- (h) The photosynthetic pigments of the leaves from the two varieties of plants were extracted and were separated by two-way chromatography. The pigments were first separated by one solvent and then separated again by a second solvent at right angles to the first solvent. Fig. 1.3 shows the results for the two different varieties.

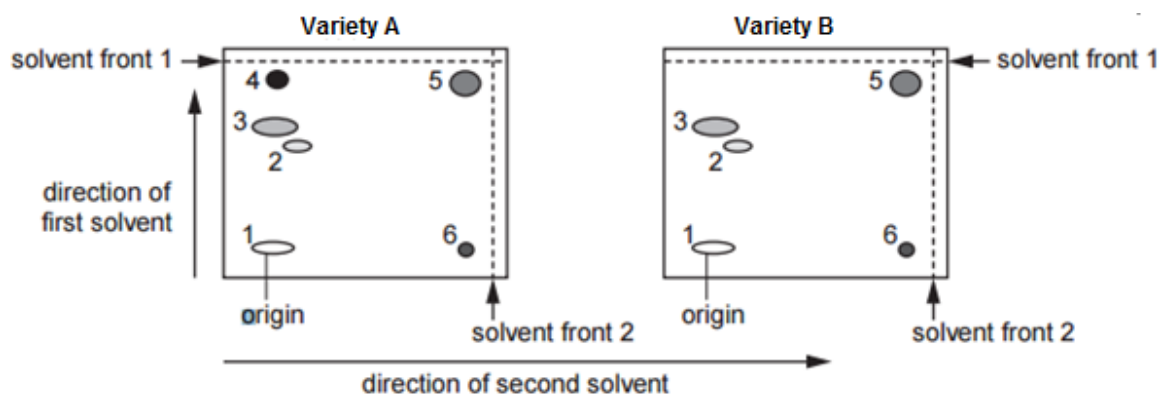


Fig. 1.3

Different photosynthetic pigments absorb different wavelengths of light. Table 1.3 shows some information about the pigments, P, Q, R, S and T, found in the 2 varieties, including the wavelength of light at which maximum light absorption occurs.

Table 1.3

pigment	wavelength of light / nm	Rf value	
		solvent 1	solvent 2
P	620	0.20	0.89
Q	545 and 547	0.60	0.29
R	420 and 660	0.65	0.11
S	490	0.91	0.19
T	430 and 645	0.82	0.92

$$R_f = \frac{\text{distance moved by pigment}}{\text{distance moved by solvent front}}$$

One of the varieties lacks one of the pigments.
Using the information in Table 1.3 and Fig. 1.3:

- (i) identify the variety that lacks one of these pigments and state the letter of the missing pigment. [1]

- (ii) state the evidence that supports your answer to (i). [2]

[Total: 21]

- 2 Methylene blue stains dead cells blue.
Living cells are not stained blue so they will appear white or clear.

You are provided with:

- methylene blue solution, M, (handle carefully as it will stain your skin)
- suspensions of yeast cells, labelled S1, S2 and S3.

Each suspension, S1, S2 and S3 has been heated for ten minutes at 45°C or 80°C or 100°C.

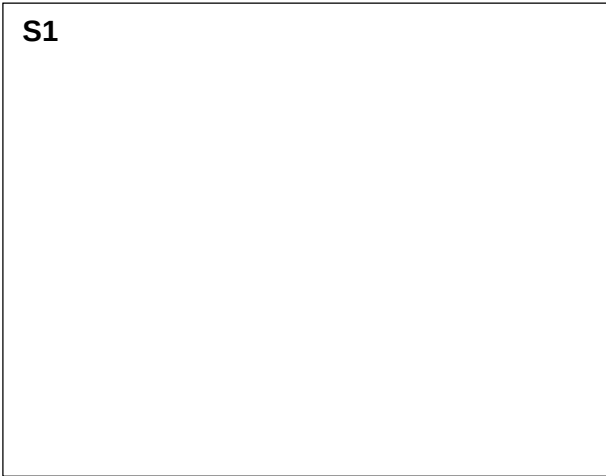
You are required to:

- use the microscope to observe the colour of the yeast cells from S1, S2 and S3, after M has been added
- record your observations by using annotated drawings of three yeast cells from each of S1, S2 and S3
- identify the temperature at which each of S1, S2 and S3 was heated.

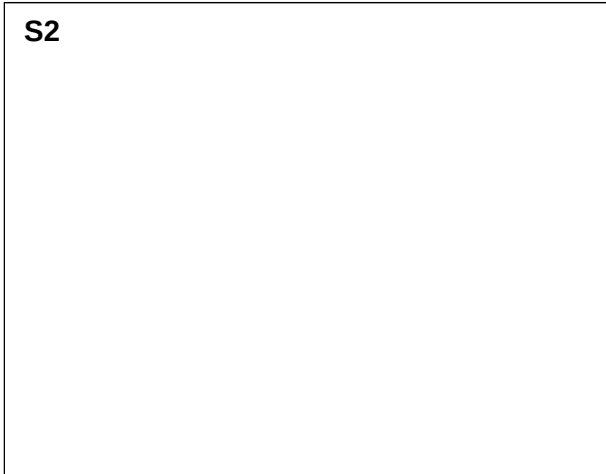
1. Label three microscope slides S1, S2 and S3.
2. Place one drop of S1 onto slide S1 and add one drop of M. Mix carefully using a glass rod. (If M comes into contact with your skin rinse with cold water.)
3. Repeat step 2 with S2 and S3.
4. Leave for five minutes.
5. Add a coverslip to each slide.
6. Use the paper towel to dry off any excess liquid around the coverslip.
7. Use the microscope to observe the yeast cells on each slide, then select cells which you can draw and annotate to describe the effect of the methylene blue, M.

- (a) (i)** Prepare the space below and record your observations by:
- making drawings of three cells from each of the slides in the boxes provided
 - annotating your drawings to describe the effect of methylene blue, M on the cells. [4]

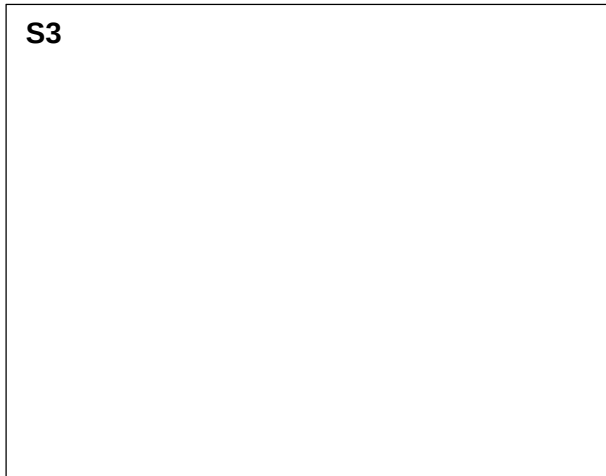
S1



S2



S3



- (ii) Use your observations to identify the temperature that was used to heat each of the suspensions S1, S2 and S3.

Complete the table. [1]

suspension	temperature / °C
S1	
S2	
S3	

- (iii) Explain how you identified the yeast cells that had been heated at 100 °C. [1]

- (b) Using the eyepiece graticule fitted in the eyepiece lens of your microscope, and the stage micrometer, find the actual length, in μm , of one of the yeast cells that you have drawn in S2.

Show the measurements that you made and your working. [3]

Actual length of a yeast cell = μm

- (c) Draw a straight line on your drawing across the yeast cell to show where you took your measurement. [1]

Use your knowledge of the actual size of the yeast cell to calculate the magnification of your drawing. [1]

- (d) The yeast *Rhodotorula glutinis* produces an enzyme, α -arabinofuranosidase, that could be used in the production of compounds to enhance the flavour and smell of fruit juices. The effect of the initial pH of the culture medium on the growth rate of this yeast was tested. Three continuous culture systems were set up, each with a different initial pH. The cultures were sampled at hourly intervals for 20 hours at each pH. The mean growth rate was then calculated.

The mean growth rates with their standard deviations are shown in Table 2.1.

Table 2.1

pH	mean growth rate / arbitrary units h ⁻¹
4.0	0.156 $\pm$ 0.001
5.2	0.197 $\pm$ 0.013
7.0	0.037 $\pm$ 0.011

A *t*-test was carried out on the results for pH 4.0 and pH 5.2 and gave the value,

$$t = 2.4$$

The degree of freedom is 38.

Based on the findings of the *t*-test, a student concluded that pH 5.2 was optimum for the production of the enzyme α -arabinofuranosidase by *R. glutinis*. Suggest two reasons why this conclusion may not be valid. [2]

- (e) A student carried out t -test on the results to compare the lengths of yeast cells when grown in different media.

A number of t -test was carried out to find out if, after 70 minutes, the difference in mean yeast cell length is significant:

1. between medium A and medium B $t = 2.50$
2. between medium A and medium C $t = 3.56$
3. between medium B and medium C $t = 1.94$

Table 2.2 shows the critical values for the t -test.

The number of degrees of freedom is 18.

Table 2.2

degrees of freedom	10	12	14	16	18	20	22	24	26	28	30	40	50	60
probability 0.05	2.23	2.18	2.14	2.12	2.10	2.09	2.07	2.06	2.06	2.05	2.04	2.02	2.01	2.00
probability 0.01	3.17	3.06	2.98	2.92	2.88	2.85	2.82	2.80	2.78	2.76	2.75	2.70	2.68	2.66

State what conclusions can be drawn about the significance of the differences in mean lengths from the three values of t given above. [3]

- (f) Fig. 2.1 is a photomicrograph of a stained transverse section through part of a plant leaf. This plant species is native to part of Asia.

You are not expected to have studied this leaf.



Fig. 2.1

Draw a large plan diagram of the part of the leaf shown in Fig. 2.1. On your diagram, use a ruled label line and label to show the vascular bundle. [4]

[Total: 20]

- 3 A number of plant tissues are coloured because the cells contain chemicals called betacyanins.

You are provided with beetroot which contains betacyanins which colour the beetroot red.

In this experiment, you will test the effect of two different alcohols – methanol and ethanol on beetroot membranes. Ethanol is found in alcoholic beverages. Methanol, sometimes referred to as wood alcohol, can cause blindness and death.

If beet membranes are damaged, the red pigment will leak out into the surrounding environment. The intensity of color in the environment should be proportional to the amount of cellular damage sustained by the beet.

Plan an investigation to find out whether or not betacyanin leakage for beetroot occurs at the same intensity using ethanol and methanol.

You must use:

- beetroot
- 40% ethanol
- 40% methanol
- colourimeter and cuvette

You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware e.g. test-tubes, boiling tubes, beakers, measuring cylinders, graduated pipettes, glass rods etc.
- stopwatch
- distilled water
- white tile
- scalpel
- forceps

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- be illustrated by relevant diagram(s), if necessary, to show, for example, the arrangement of the apparatus used
- identify the independent and dependent variables
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible
- include layout of results tables and graphs with clear headings and labels
- use the correct technical and scientific terms
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 14]

[illegible]

[illegible]